

MAT 523: Functions of a Real Variable I

Homework I

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1.

Show that $\mathbb{C}$ (the field of complex numbers) is not an ordered field.

Solution

Lemma 1A: $(-1)(-1) = (-1)^2 = 1$.

Proof:

$$\begin{aligned}(-1)(-1) &= (-1)(-1) + 0 \\&= (-1)(-1) + [1 + (-1)] \\&= [(-1)(-1) + (-1)] + 1 \\&= [(-1)(-1) + (-1)(1)] + 1 \\&= (-1)[-1 + 1] + 1 \\&= (-1)(0) + 1 \\&= 0 + 1 \\&= 1.\end{aligned}$$

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Lemma 1B: Let F be a field and let $a \in F$. Then $-a = (-1)a$.

Proof:

$$\begin{aligned}(-1)a &= (-1)a + 0 \\&= (-1)a + [a + (-a)] \\&= [(-1)a + a] + (-a) \\&= [a(-1) + a(1)] + (-a) \\&= a[(-1) + 1] + (-a) \\&= a[0] + (-a) \\&= 0 + (-a) \\&= -a.\end{aligned}$$

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Lemma 1C: $1 > 0$.

Proof: By the Non-triviality Assumption, $1 \neq 0$, so suppose $1 < 0$. Then $0 - 1$ is positive, and by the first positivity axiom $(0 - 1)(0 - 1)$ is positive. But

$$(0 - 1)(0 - 1) = [0 + (-1)][0 + (-1)] \quad (1)$$

$$= [(-1) + 0][(-1) + 0] \quad (2)$$

$$= (-1)(-1) \quad (3)$$

$$= 1 \quad (4)$$

by Lemma 1A. By assumption $1 < 0$, so $(0 - 1)(0 - 1) < 0$. This contradicts the fact that $(0 - 1)(0 - 1)$ is positive. Therefore, $1 > 0$.

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Corollary 1A: $-1 < 0$.

Proof: Since $1 > 0$ by Lemma 1C, it would be a violation of the second positivity axiom if $-1 > 0$ as well. Therefore, $-1 < 0$.

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Theorem: $\mathbb{C}$ is not an ordered field.

Proof: Suppose $\mathbb{C}$ is an ordered field. Then an order $<$ can be defined on $\mathbb{C}$. Consider $i, 0 \in \mathbb{C}$. If $0 < i$, then, by the first positivity axiom, it must be the case that $0 < ii = i^2 = -1$. By Corollary 1A, $-1 < 0$, so we have a contradiction. If, on the other hand, $i < 0$, then $0 < -i$, by the second positivity axiom. By the first positivity axiom, it follows that

$$\begin{aligned} 0 &< (-i)(-i) \\ &= [(-1)i][(-1)i] \quad (\text{by Lemma 1B}) \\ &= [(-1)(-1)][ii] \\ &= 1i^2 \quad (\text{by Lemma 1A}) \\ &= i^2 \\ &= -1. \end{aligned}$$

Again, this is a contradiction. Therefore, $\mathbb{C}$ is not an ordered field.

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Verify the following:

- (i) For each real number $a \neq 0$, $a^2 > 0$. In particular, $1 > 0$, since $1 \neq 0$ and $1 = 1^2$.
- (ii) For each positive number a , its multiplicative inverse a^{-1} also is positive.
- (iii) If $a > b$, then $ac > bc$ if $c > 0$ and $ac < bc$ if $c < 0$.

Solution

(i) **Proof:** Let $a \in \mathbb{R} \setminus \{0\}$. If $a > 0$, then $aa = a^2 > 0$, by the first positivity axiom. If $a < 0$, then $0 < -a$.

Thus,

$$\begin{aligned}
 0 &< (-a)(-a) \\
 &= [(-1)a][(-1)a] \quad (\text{by Lemma 1B}) \\
 &= [(-1)(-1)][aa] \\
 &= 1a^2 \quad (\text{by Lemma 1A}) \\
 &= a^2.
 \end{aligned}$$

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(ii) **Proof:** Let $a \in \mathbb{R}^+$. Suppose $a^{-1} < 0$. Then $0 < -a^{-1}$. Thus,

$$\begin{aligned}
 0 &< (-a^{-1})(a) \\
 &= [(-1)a^{-1}]a \quad (\text{by Lemma 1B}) \\
 &= (-1)[aa^{-1}] \\
 &= (-1)1 \\
 &= -1 \quad (\text{by Lemma 1B})
 \end{aligned}$$

By Corollary 1A, $-1 < 0$, so we have a contradiction. Therefore, $a^{-1} > 0$.

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(iii) Let $a, b, c \in \mathbb{R}$ such that $a > b$. Since $a > b$, it follows that $a - b > 0$. Suppose $c > 0$. Then, by the first positivity axiom, $c(a - b) > 0$. But

$$\begin{aligned}
c(a - b) &= c[a + (-b)] \\
&= ca + c(-b) \\
&= ac + (-b)c \\
&= ac + [(-1)b]c && \text{(by Lemma 1B)} \\
&= ac + (-1)[bc] \\
&= ac - bc.
\end{aligned}$$

Thus, $ac - bc > 0$, or $ac > bc$.

Now suppose $c < 0$. Then $0 < -c$, so $(-c)(a - b) > 0$. But

$$\begin{aligned}
(-c)(a - b) &= [(-1)c](a + (-b)) \\
&= (-1)[c(a + (-b))] \\
&= (-1)[ca + c(-b)] \\
&= (-1)[ca + c[(-1)b]] && \text{(by Lemma 1B)} \\
&= (-1)[ac + (-1)[cb]] \\
&= (-1)(ac) + [(-1)(-1)](bc) \\
&= (-1)(ac) + 1(bc) && \text{(by Lemma 1A)} \\
&= bc + (-1)(ac) \\
&= bc - ac.
\end{aligned}$$

Therefore, $bc - ac > 0$ or $bc > ac$.

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3.

Let $E \subseteq \mathbb{R}$ be a nonempty subset that is bounded below. Show that E has a greatest lower bound and

$$\inf E = -\sup\{-x|x \in E\}.$$

Proof

Let $A = \{-x|x \in E\}$. Since E is bounded below, A must be bounded above. Thus, by the Completeness Axiom, A has a least upper bound $\sup A$. Since $\sup A$ is an upper bound for A , $-\sup A$ must be a lower bound for E . We show that it is the greatest lower bound for E . If not, then there is a lower bound ℓ for E with $\ell > -\sup A$. If ℓ is a lower bound for E , then $-\ell$ is an upper bound for A and satisfies $-\ell < \sup A$. But this is a contradiction, since there can be no upper bound less than the supremum. Therefore, $-\sup A$ is the greatest lower bound for E , i.e.,

$$\inf E = -\sup A.$$

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5.

Let $E \subset \mathbb{R}$ and suppose $g \in \mathbb{R}$ is an upper bound for E . Prove that g is the least upper bound for E if and only if for all $\epsilon > 0$ $E \cap [g - \epsilon, g] \neq \emptyset$, i.e., there exists $x \in E$ such that

$$g - \epsilon \leq x \leq g.$$

Proof

First, suppose $g = \sup E$, but there exists an $\epsilon > 0$ such that $E \cap [g - \epsilon, g] = \emptyset$. Then all $x \in E$ satisfy either $x < g - \epsilon$ or $x > g$. Clearly, no $x \in E$ can satisfy $x > g$, since g is an upper bound for E . Therefore, $x < g - \epsilon$ for all $x \in E$. This makes $g - \epsilon$ an upper bound for E ; however, $g - \epsilon$ is clearly smaller than g , which contradicts the fact that g is the supremum of E .

Next, if $E \cap [g - \epsilon, g] \neq \emptyset$ for arbitrary $\epsilon > 0$, it follows that there exists an $x \in E$ such that $g - \epsilon \leq x \leq g$ for all $\epsilon > 0$. This means that no matter how small we choose ϵ to be, there will always be an element of E between $g - \epsilon$ and g . Clearly, this means that no number smaller than g can be an upper bound for E . Therefore, g is the supremum of E .

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7.

Let A be a bounded subset of $\mathbb{R}$. Let $B \subset \mathbb{R}$ be the set of upper bounds for A . Then $\sup A = \inf B$.

Solution

Lemma 7A: The set of upper bounds for the empty set is all of $\mathbb{R}$.

Proof: Let $b \in \mathbb{R}$. The statement

$$a \in \emptyset \text{ implies } a \leq b$$

is true because the antecedent is false. Therefore, b is an upper bound for $\emptyset$. Since b was an arbitrary real number, we conclude that the set of upper bounds of the empty set is all of $\mathbb{R}$. ■

Theorem: Let A be a bounded subset of $\mathbb{R}$. Let $B \subset \mathbb{R}$ be the set of upper bounds for A . Then $\sup A = \inf B$.

Proof: If $A = \emptyset$, then $\sup A = -\infty$. By Lemma 7A, the set of upper bounds for A is $B = \mathbb{R}$. The infimum of $\mathbb{R}$ is $\inf \mathbb{R} = -\infty$. So we have $\sup A = \inf B$. Next, suppose $A \neq \emptyset$. Since A is bounded, it is bounded above. By the Completeness Axiom, A must, therefore, have a supremum $\sup A = b_0$. Since b_0 is the supremum of A , it follows that $b_0 < b$ for all other upper bounds b of A . That is, $b_0 < b$ for all $b \in B \setminus \{b_0\}$. This further implies that B is bounded below and b_0 is a lower bound for B . Since B is bounded below, it must have an infimum $\inf B$. We show that $\inf B = b_0$. Let β be a lower bound for B . Suppose $\beta > b_0$. Then β is not a lower bound for B since $b_0 \in B$ and $b_0 < \beta$. This is a contradiction. Therefore, $b_0 = \inf B = \sup A$. ■